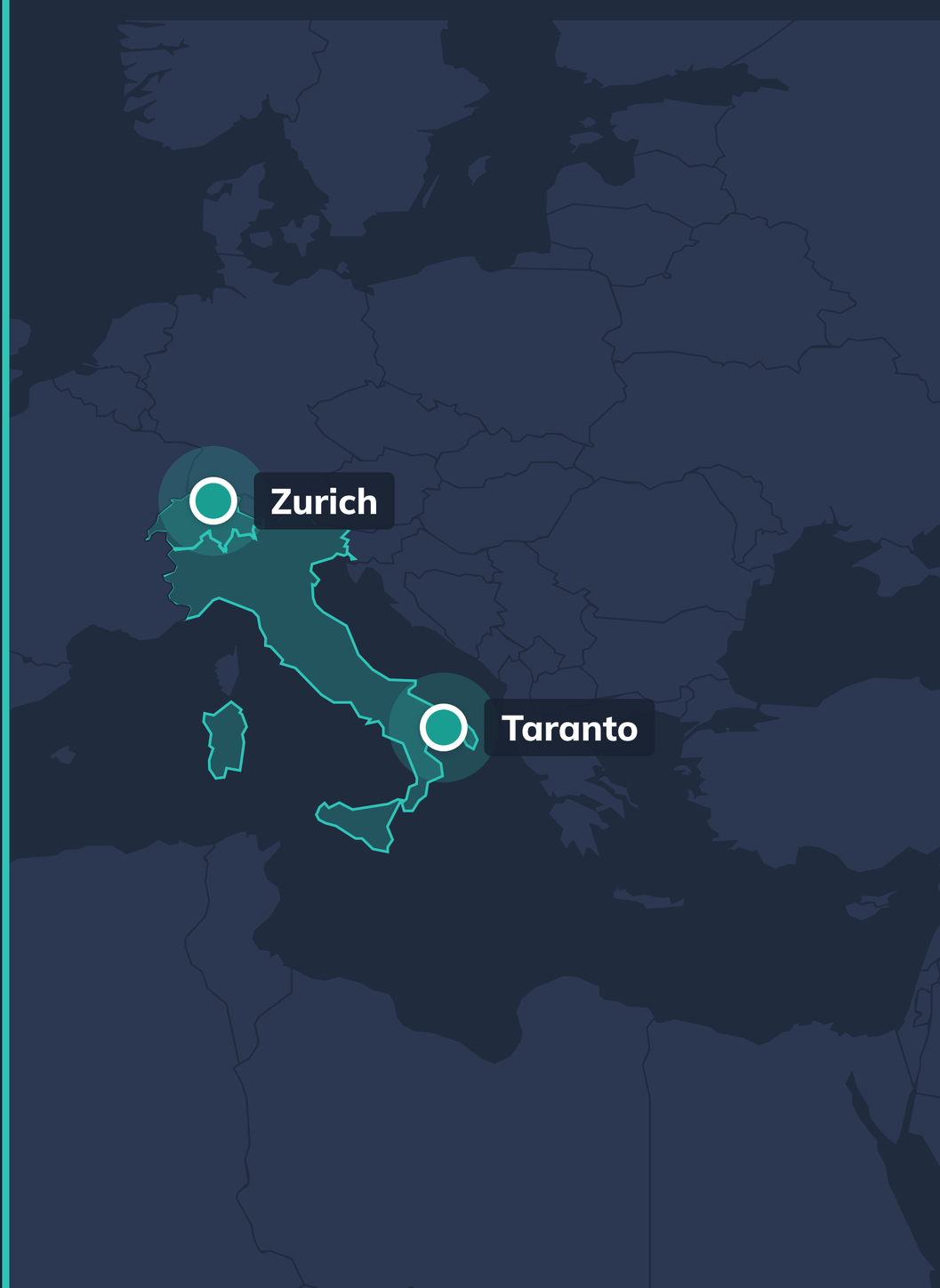
KAUST SCALEX 2026

bioergotech

We build software for hospitals, doctors and pharmacists.

Each product starts from a specific problem in a hospital, a health authority or a medical society.



Guido Putignano, Founder and President

Zurich and Taranto, info@bioergotech.org

What we do

We are engineers, biostatisticians, clinicians and lawyers, working between Zurich and Taranto, with experience at ETH Zurich, Harvard and MIT. We work with hospitals, health authorities and medical societies on a problem they already have, often first as a paid consulting project.

When a solution works, we turn it into a product that other institutions can buy.

2025
We started in April

4
Products so far

3
Countries: Italy, Switzerland, the US

Frankura

For oncology departments.

With ASL VCO, in Piedmont.

• PILOT AWAITING APPROVAL

VIS Pharma Compass

For hospital pharmacies and regions.

Tested on data from Regione Abruzzo.

• PROTOTYPE

Bonnie

For rheumatology patients.

Designed with the Italian College of Rheumatologists.

• INTERFACE PROTOTYPE

Skippy

For people leaving prison.

With the University of Maryland, Baltimore.

• INTERFACE PROTOTYPE

Team

Foundation board: Guido Putignano, Domenica Leone and Carmine Pisano.



Guido Putignano
Founder and President

Biomedical engineer, with experience at ETH Zurich, Harvard and MIT.



Domenica Leone
COO

Lawyer. Runs operations, legal and compliance.



Olufemi Olusola
Technical lead

Biostatistician with 10 years of experience.



Saria Miccoli
Communication lead

Designer. Shapes our visual identity and communications.



Mario Tagarelli
Economic manager

Looks after finance, grants and reporting.

ADVISORS



Daniela Marotto
Rheumatologist, Italian College of Rheumatologists (CReI)



Pasquale Persico
Market access at Gilead Sciences



Roberto De Ponti
Managing Director, 3B Future Health Fund

Who we work with

Doctors

Oncologists and rheumatologists. They lose hours to paperwork and see chronic patients only every few months.

- FRANKURA AND BONNIE

Pharmacists

Hospital pharmacy directors and regional pharmacy services. They manage large drug budgets, often in spreadsheets.

- VIS PHARMA COMPASS

Hospitals and regions

Local health authorities, hospitals and regional governments. They hold the budget, so they are the ones who sign the contract.

- ALL PRODUCTS

Medical societies and universities are partners in some projects. The Italian College of Rheumatologists is working with us on the design of the Bonnie study, and the University of Maryland, Baltimore runs Skippy with us.

Frankura

PILOT AWAITING APPROVAL

For oncology departments. Built with ASL VCO, in Piedmont. The pilot is waiting on regional approval.

ESTIMATED DOCTOR TIME SAVED PER YEAR

About

1,400

hours a year

CLOSE TO

€67,000

at €48 an hour

ROUGHLY

seven months

of one doctor's full-time work

Our estimate, based on ASL VCO's own volumes.

EVERY YEAR ASL VCO HANDLES

1,100
cancer intake forms

525
oncology visits

1,547
tumour board cases

Most of the data is typed by hand and copied from one form to the next.

WHAT FRANKURA DOES	TIME SAVED
Records the visit and fills in the intake form	406 h
Copies the data into the tumour board file	600 h
Prepares a one-page summary of each case for the board, with the options suggested by the AIOM and ESMO guidelines. The board confirms, changes or rejects them.	386 h
Produces a standard report for GPs and other hospitals	

It runs on top of the hospital's existing software.

How big the market is for Frankura

AI AGENTS IN HEALTHCARE

$6.9bn

The global market by 2030, up from $1.1 billion in 2025.



The nearby market for AI-powered clinical documentation reaches about $14 billion in the same year.

PUBLIC BUYERS

218

The public institutions that buy software like ours: 198 health authorities and hospital trusts in Italy, and 20 health clusters in Saudi Arabia.

At €50,000 to €150,000 per licence, that is **€11 to €33 million**, plus a yearly fee.

THREE-YEAR TARGET

20 hospitals

Our target for the next three years, worth **€1 to €3 million** in licences.

Four Italian institutions are already talking to us: ASL Salerno, Ospedale Miulli, ASL Taranto and ASL Caserta.

Sources: MarketsandMarkets 2026, The Business Research Company 2026, CERGAS Bocconi OASI Report 2025, Health Holding Company.

VIS Pharma Compass

PROTOTYPE

For hospital pharmacies and regions. A working prototype, not yet under contract.

€4.8bn spent above the legal ceiling in 2025 on the medicines the Italian public health system buys directly, up from €4.0 billion in 2024 (AIFA).

Pharmacy teams have the data but work in spreadsheets, so they spot savings late.

WHAT VIS DOES

- Puts purchase and dispensing data in one place and checks it against the accounts
- Flags patents about to expire and biosimilars that are available but not used
- Compares health authorities in the same region, medicine by medicine

It only reads aggregated data, so it is not a medical device.

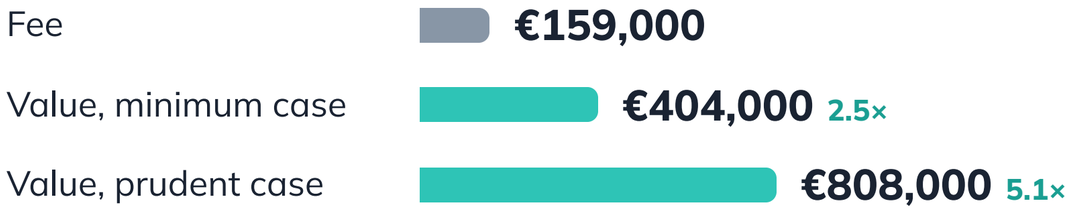
Tested on Regione Abruzzo's four health authorities

250,000+ dispensing records

about €450M of yearly spend

YEARLY FEE AND ESTIMATED YEARLY VALUE

For a region with four health authorities



€9,000 Sales start with an analysis of the client's own data, done in three weeks.

Bonnie

INTERFACE PROTOTYPE

For rheumatology patients, designed with the Italian College of Rheumatologists (CReI).

A patient with rheumatoid or psoriatic arthritis sees the rheumatologist every three to six months. Flares, side effects and stopped treatments in between are rarely written down, so each visit starts from what the patient remembers.

HOW IT WORKS

- Every week the patient answers a few questions in about two minutes, with no account to create
- Bonnie, a small cartoon skeleton, reminds them and keeps them going. It reacts to the fact that they answered, never to what they said
- Before the visit, the rheumatologist sees how the patient has been, week by week

THE STUDY WITH CREI

500 About 500 rheumatologists, in hospitals and in the community, belong to CReI.

We are designing the study with them. Called PROSSIMA, it would follow patients with

RHEUMATOID ARTHRITIS · PSORIATIC ARTHRITIS · AXIAL SPONDYLOARTHRITIS

CReI would own the study and the data. We build and run the software. We built the prototype at our own cost before asking them to commit.

Bonnie is built on the Frankura platform. It is not a medical device.

Skippy

INTERFACE PROTOTYPE

For people leaving prison, with the EMBRACE Initiative at the University of Maryland, Baltimore.

People coming out of prison are usually handed a list of phone numbers. Programmes count a referral as soon as it is sent, even if the person never gets help.

HOW IT WORKS

- Helps the person take the next practical step, such as getting an ID or applying for benefits
- Connects them with the right service or staff member
- Follows up until the step is done, and tells staff what got in the way

THE PLAN WITH EMBRACE



EMBRACE owns the programme and the data. We build and run the software. It is our first project in the United States.

How we work

1

Consulting first

A hospital, region or medical society comes to us with a problem. We work on it as a paid project, or start with a paid analysis like the €9,000 one for VIS. We build a first prototype at our own cost, and the full product only once someone is paying for the work.

2

One platform

What we build goes into one shared platform, Frankura: recording clinical data, patient consent, audit trails, the AI layer and data storage in Europe. New products start from there, and Bonnie is built on it.

3

Sell it again

Once a product works for one client, we offer it to the next. ASL VCO follows the intake and tumour board model of the Piedmont oncology network, so the same workflow applies across the region. A new region for VIS needs a data adapter, not a new system.

bioERGOtech Foundation
Finds the clinical needs, and does the research and consulting.

bioERGOtech clinical software
A dedicated company, which we are setting up, will sell and run the software in hospitals.

Where we are in 2026

1 pilot

With ASL VCO for Frankura, waiting on regional approval.

4 health authorities

VIS tested on real data from Regione Abruzzo.

4 in talks

ASL Salerno, Ospedale Miulli, ASL Taranto and ASL Caserta.

15 contacts

Health authorities, hospitals and regions in Italy that we can reach directly.



Pilot awaiting approval · Tested on data · In talks

Map: ISTAT boundaries, CC BY 4.0.

The research behind the products

Each project is listed on bioergotech.org, with what it has shown and what it has not.

Cancer risk after inflammatory disease

PUBLISHED

In *Cancers*, 2026, with Daniela Marotto. A national cohort of 356,022 patients from Italian hospital discharge records. Cancer risk is highest in the first year after diagnosis.

Cancer risk by subgroup

COMPLETED

An algorithm that links genotype, lifestyle and socioeconomic data to cancer incidence at population level. Delivered within CODIGE, part of the AGE-IT programme.

Cell-therapy models

PUBLISHED

Mathematical models of engineered cell therapies, at ETH Zurich. Review published in *Frontiers in Immunology*, 2025.

Very small patient groups

IN PROGRESS

Methods for diseases too rare for standard statistics, starting with propionic acidemia in Saudi Arabia.

Guido Putignano works at ETH Zurich's Department of Biosystems Science and Engineering. Our lab is in Taranto.

OncoTarget

GRANT APPLICATION SUBMITTED

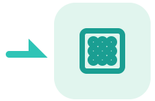
Testing cancer drugs on organoids grown from a patient's own tumour. A gastric cancer project at the University of Zurich, led by Giovanni Papa.

Patients with gastric cancer can respond very differently to the same treatment. Genetic profiling finds relevant changes, but it does not always show which available drug will work for a given patient.

HOW IT WORKS



A tumour biopsy is grown into organoids, small 3D models of the patient's tumour



Automated screening tests them against a panel of cancer drugs used in the clinic



The drug responses are read together with the tumour's molecular profile

OncoTarget and Frankura are separate projects. They could meet in future at the tumour board, where treatment is chosen.

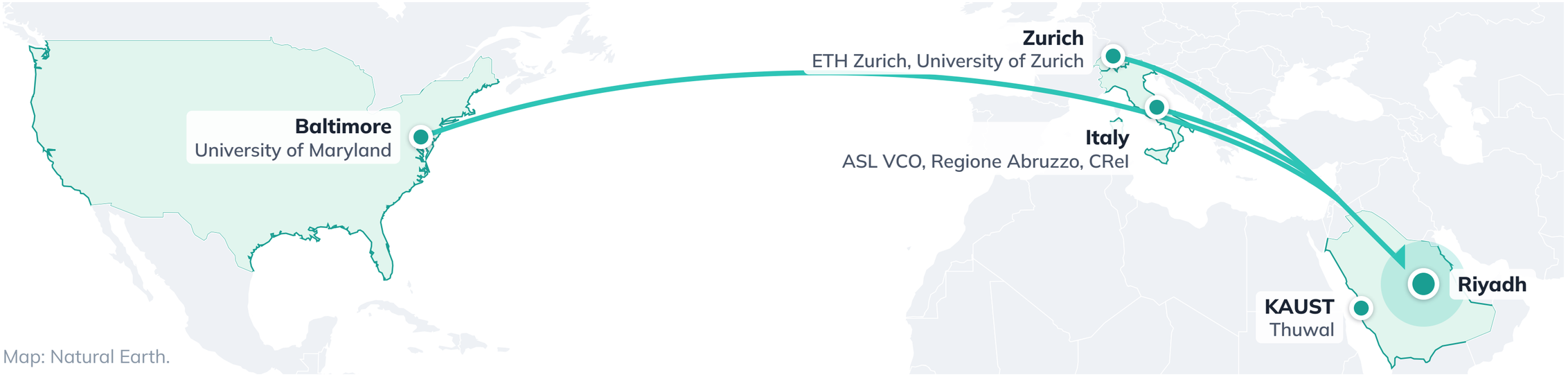
WHERE IT STANDS

- Protocols for human gastric organoids, with the Departments of Visceral Surgery and Pathology at the University Hospital Zurich
- Drug screening in 3D organoids, with the Children's Hospital Zurich
- Drug responses already linked to specific genetic backgrounds, with most hits validated one by one

Giovanni Papa (UZH, Institute of Molecular Cancer Research) leads, with Prof. Anne Müller as mentor. Guido Putignano builds the data model, the analysis pipeline and the quality criteria.

How we get to Saudi Arabia

We test each product where it is easiest to prove, keep what works, and offer that to Saudi health clusters.



Map: Natural Earth.

United States

Does it work for thousands of people? With the University of Maryland, Baltimore.

Switzerland

Does the science hold up? With ETH Zurich and the University of Zurich.

Italy

Do public hospitals buy it and use it? With ASL VCO, Regione Abruzzo and CReI.

Saudi Arabia

We offer what passed to health clusters, starting with one pilot through ScaleX.

Our plan for Saudi Arabia

WHAT WE KEEP FROM EACH TEST

- Products with a paying customer
- Results we have measured
- Nothing that needs custom work for each new client

HOW WE REACH PROFIT IN SAUDI ARABIA

Paid analyses of each cluster's own data first, as we do with VIS in Italy. Then licences and yearly fees. We aim to be profitable in Saudi Arabia by the end of 2027.

2026

Testing in Italy, the US and Switzerland

2027

ScaleX, a first pilot and first contract. Profitable in Saudi Arabia by the end of the year

2028

Five Saudi health clusters

2029

Ten clusters, half of the Kingdom

2030

All 20 clusters, over 20 million people

Who this reaches

24 million

Our goal for 2030: our products in use in all 20 Saudi health clusters and in the Italian health systems we already work with, together about 24 million people.



ASL VCO, Frankura	**153,000**	PILOT AWAITING APPROVAL
Regione Abruzzo, VIS	**1.27 million**	TESTED ON DATA
ASL Salerno, Taranto and Caserta	**2.5 million**	IN TALKS
All 20 Saudi health clusters	**Over 20 million**	GOAL FOR 2030
Total	**About 24 million**	

Residents on 1 January 2025 (ISTAT). The 20 Saudi health clusters serve over 20 million people (Health Holding Company).

Skippy adds 2,000 people leaving prison in Maryland.

What we are asking from ScaleX

We want to open in Riyadh with products that are already being tested in Italian public health. Saudi public healthcare is moving to 20 regional health clusters, each running the hospitals and clinics for about 1 million people. That is close to how the Italian health authorities we work with are organised.

WHAT WE WOULD BRING

- **Frankura**, for oncology departments
- **VIS Pharma Compass**, for hospital drug budgets
- **Bonnie**, for patients with chronic conditions between visits

WHAT WE NEED

- An introduction to a health cluster for a first pilot
- Advice on Saudi rules for clinical software and health data
- Advice on the fastest route to a profitable Saudi operation
- Introductions to investors
- Lab access at KAUST and a desk in Riyadh for our team

A first pilot would follow our Italian path

3 weeks → A paid analysis of the cluster's own data

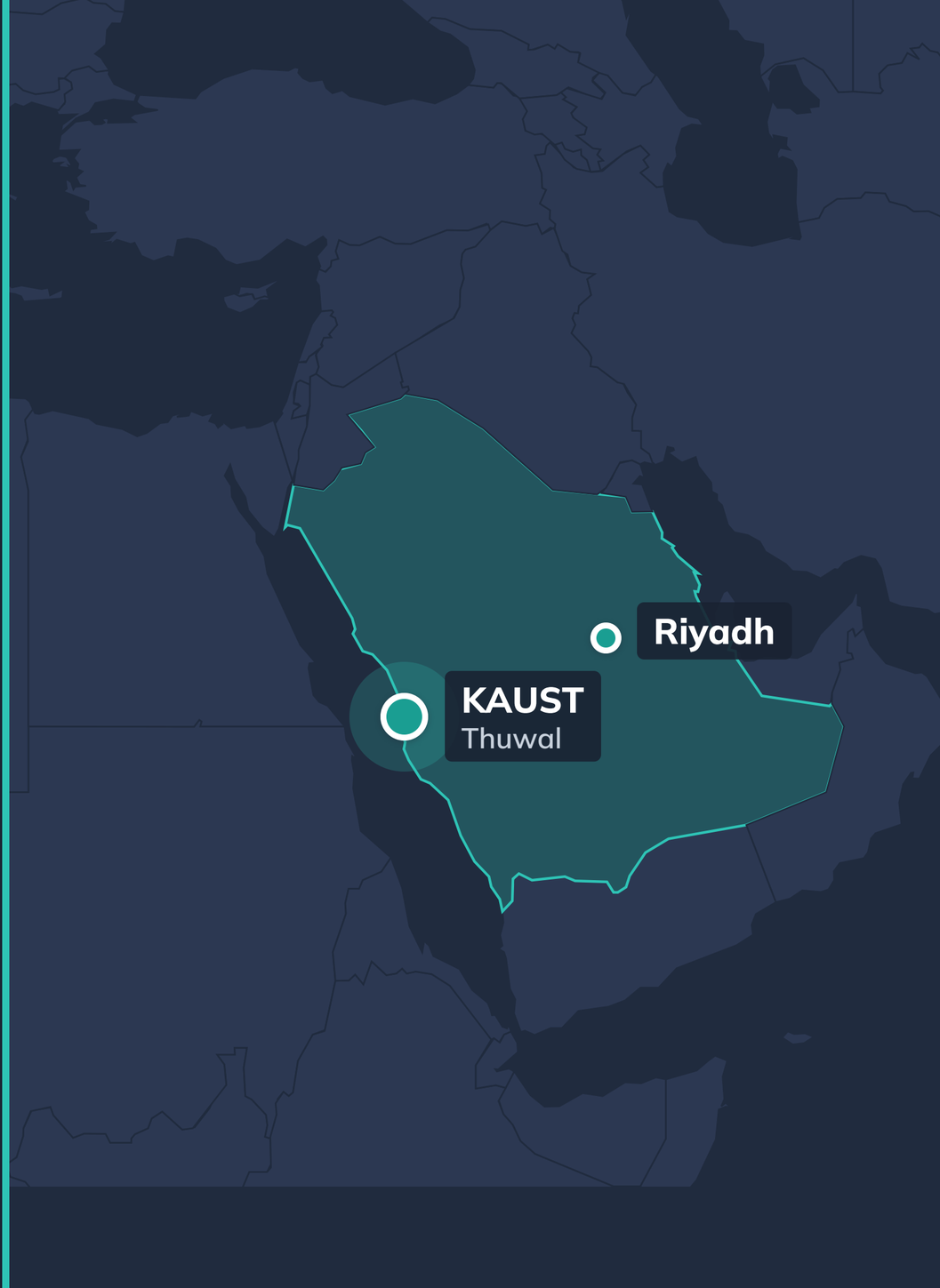
90 days → The software running

6 months → A joint review

bioergotech

Thank you

Four products, a pilot with ASL VCO awaiting regional approval, and a plan to reach all 20 Saudi health clusters by 2030.



Guido Putignano, Founder and President

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